

EXPERIMENT NO. : 1 (ASSIGNMENT)DATE: 13/08/2026

1. AIM

To design and study a lithography mask pattern for shallow trench isolation, illustrating line space structures within a defined boundary for device isolation for given masks.

2. TOOL

Semulator3D

3. THEORY

Shallow Trench Isolation (STI) is a semiconductor fabrication technique used to electrically isolate adjacent devices. In this process, shallow trenches are formed in the silicon substrate and filled with an insulating material. Lithography is used to define the regions where the trenches are to be formed.

In this assignment, two lithography mask designs are studied. Design 1 consists of a $500\text{ nm} \times 400\text{ nm}$ boundary containing four vertical mask openings, each 40 nm wide and separated by 60 nm spacing. The openings form a regular line-space grating oriented along the Y-axis. Design 2 consists of a $500\text{ nm} \times 500\text{ nm}$ boundary with a 60 nm wide horizontal bar and a 60 nm wide vertical bar, both extending across the complete boundary and intersecting at the center.

These patterns illustrate how lithographic dimensions such as opening width, spacing, orientation, and alignment are used to define regions for device isolation. Precise control of these parameters is important for transferring the required pattern accurately during semiconductor fabrication.

4. DESIGN PARAMETERS

Table 1: assignment 1

Number	Lock	Action	Name	Material(s)	Dopant(s)	Depth	Mask	Mask Polarity	Wafer	Wafer Side	Lateral Ratio	Parameter Name	Comment
1		Wafer Setup	Wafer Setup	Silicon/Silicon...		200			WAFER_1				
2		Conformal...	pad oxide deposition	/Oxides/Silicon...		4			WAFER_1	Both	1.0		
3		Conformal...	pad nitride deposition	/Nitrides/SiN		40			WAFER_1	Top	1.0		
4		Sequence	lithography						WAFER_1				
4.1		Planarizing...	spin on resist	/Organics/photo...		120			WAFER_1	Top			
4.2		Expose resist	Expose resist	/Organics/photo...		0	tx	Light	WAFER_1	Top			
5		Sequence	etch fins						WAFER_1				
5.1		Enhanced B...	pad nitride etch	/Nitrides/Org...		40			WAFER_1	Top	0.05		
5.2		Enhanced B...	pad oxide etch	/Nitrides/Org...		4			WAFER_1	Top	0.05		
5.3		Enhanced B...	silicon etch	/Nitrides/Org...		100			WAFER_1	Top	0.05		
6		Remove M...	resist strip	/Organics/					WAFER_1				
7		Conformal...	sti fill deposition	/Oxides/SiO2_H...		200			WAFER_1	Top	0.67		
8		Polish	sti oxide cmp						WAFER_1	Top			

Table 2: assignment 2

Number	Lock	Action	Name	Material(s)	Dopant(s)	Depth	Mask	Mask Polarity	Wafer	Wafer Side	Lateral Ratio	Parameter Name	Comment
1		Wafer Setup	Wafer Setup	Silicon/Silicon...		200			WAFER_1				
2		Conformal...	pad oxide deposition	/Oxides/Silicon...		4			WAFER_1	Both	1.0		
3		Conformal...	pad nitride deposition	/Nitrides/SiN		40			WAFER_1	Top	1.0		
4		Sequence	lithography						WAFER_1				
4.1		Planarizing...	spin on resist	/Organics/photo...		120			WAFER_1	Top			
4.2		Expose resist	Expose resist	/Organics/photo...		0	cross_line	Light	WAFER_1	Top			
5		Sequence	etch fins						WAFER_1				
5.1		Enhanced B...	pad nitride etch	/Nitrides/Org...		40			WAFER_1	Top	0.05		
5.2		Enhanced B...	pad oxide etch	/Nitrides/Org...		4			WAFER_1	Top	0.05		
5.3		Enhanced B...	silicon etch	/Nitrides/Org...		100			WAFER_1	Top	0.05		
6		Remove M...	resist strip	/Organics/					WAFER_1				
7		Conformal...	sti fill deposition	/Oxides/SiO2_H...		200			WAFER_1	Top	0.67		
8		Polish	sti oxide cmp						WAFER_1	Top			

7. OBSERVATION

Design 1 showed four uniform vertical line openings within the $500\text{ nm} \times 400\text{ nm}$ boundary. Each line was 40 nm wide with 60 nm spacing, producing a regular line-space grating along the Y-axis.

Design 2 showed a cross-shaped pattern within the $500\text{ nm} \times 500\text{ nm}$ boundary. The horizontal and vertical openings were each 60 nm wide, extended across the full boundary, and intersected at the center.

8. CONCLUSION

The lithography mask patterns for shallow trench isolation were successfully designed and studied using Simulator3D. The parallel line array and orthogonal cross pattern were created with the specified dimensions, spacing, orientation, and boundary limits. The experiment demonstrated how accurately defined mask openings are used to represent isolation regions during semiconductor fabrication.

EXPERIMENT NO. : 1DATE: 13/08/2026

1. AIM

Identification of different types of Substrates

2. APPARATUS REQUIRED

Different types of substrate samples such as Sapphire, ITO, FTO, PDMS, and PET.

3. THEORY

In semiconductor fabrication and materials science, a substrate is the foundational physical base or wafer upon which thin films, active layers, or microstructures are deposited, grown, or etched.

Sapphire is a single-crystal aluminum oxide (corundum) used as an insulating, transparent, and rigid substrate.

ITO and FTO are Transparent Conducting Oxides (TCOs) that allow light to pass through while conducting electricity. They are used in optoelectronics because conventional metals are opaque. They are generally rigid, while ITO can also be coated on PET to provide flexibility.

PDMS (Polydimethylsiloxane) is a silicone-based organic polymer that is optically transparent, elastomeric, flexible, and stretchable. It is commonly used in wearable sensors.

PET (Polyethylene Terephthalate) is a transparent and flexible polymer substrate used where lightweight and bendable support is required.

5. OBSERVATION

The different substrate samples were identified based on their physical and functional properties. Sapphire was observed to be transparent, rigid, and insulating. ITO and FTO were identified as transparent conducting substrates. PDMS was identified as transparent, flexible, and stretchable, while PET was identified as transparent and flexible.

6. CONCLUSION

Different types of substrates used in semiconductor, optoelectronic, and flexible electronic applications were successfully identified. Sapphire, ITO, FTO, PDMS, and PET were distinguished based on properties such as transparency, conductivity, rigidity, flexibility, and stretchability. The experiment provided an understanding of how substrate properties determine their suitability for different device fabrication applications.

EXPERIMENT NO. : 2DATE: 13/08/2026

1. AIM

Demonstration of Processing Steps Involved in the cleaning of Glass/Silicon Wafers

2. TOOL

Glass/Silicon wafer or substrate

Ultrasonic sonicator

Beakers or sample holders

DI water

Acetone

Isopropyl alcohol (IPA)

Nitrogen blow-off or air-drying facility

PPE such as gloves and goggles

3. THEORY

Cleaning is a critical step in semiconductor fabrication because the wafer surface must be completely free from any contamination before processes such as lithography or deposition can be performed. Even tiny particles or dust can negatively impact the final device performance.

Different cleaning methods use a predefined sequence of chemicals, each selected for removing a specific type of impurity like organic residues, salts, dust, or other unwanted materials. The order and timing of these chemicals form the cleaning recipe.

During this demonstration, a glass substrate was cleaned using various chemicals along with an ultrasonic sonicator. The ultrasonic vibrations help dislodge fine particles from the surface, while the chemicals dissolve or wash away remaining contaminants. This process ensures that the substrate becomes properly cleaned, smooth, and suitable for the next fabrication steps.

4. PROCEDURE

1. Prepare the Sonicator & Samples

- Fill the sonicator bath with water (or the appropriate solvent if needed).
- Place samples in a beaker or holder (ensure they are sonicator-safe).

2. First Cleaning Cycle (DI Water)

- Submerge samples in a beaker filled with DI water.
- Run the sonicator for 15 minutes at recommended power.
- Removes loose particles and water-soluble contaminants.

3. Rinse & Inspect

- Remove samples and rinse with fresh DI water.
- Check for visible residues; repeat if needed.

4. Second Cleaning Cycle (Acetone)

- Place samples in a clean beaker with acetone (for organic residues).
- Sonicate for 15 minutes (acetone evaporates quickly).

5. Rinse & Transition to IPA

- Discard acetone, rinse samples with fresh IPA to avoid water streaks.

6. Final Cleaning Cycle (IPA)

- Sonicate samples in IPA for 15 minutes for degreasing & drying.
- Ensures no solvent residues remain.

7. Dry & Verify Cleanliness

- Remove samples, air-dry or use nitrogen blow-off.
- Inspect under a microscope if needed.

Safety Notes:

- Wear PPE (gloves, goggles).
- Work in a fume hood for acetone/IPA.
- Dispose of solvents properly.

6. OBSERVATION

The glass substrate was cleaned sequentially using DI water, acetone, and IPA with ultrasonic sonication. Loose particles, water-soluble contaminants, and organic residues were removed, and the substrate appeared clean and free from visible residues after drying.

7. CONCLUSION

The cleaning process for the glass/silicon substrate was successfully demonstrated using DI water, acetone, IPA, and ultrasonic sonication. Each cleaning stage removed specific types of contaminants and prepared the surface for further semiconductor processing. The experiment demonstrated the importance of proper wafer cleaning in achieving a clean and suitable substrate.

EXPERIMENT NO. : 3DATE: 13/08/2026

1. AIM

To design and study a lithography mask pattern for shallow trench isolation, illustrating line-space structures within a defined boundary for device isolation.

2. TOOL

Semulator3D

3. THEORY

Shallow Trench Isolation (STI) is a widely used technique in integrated circuit fabrication that provides electrical insulation between closely placed semiconductor devices so that unwanted current leakage does not occur. In this method, shallow trenches are etched into the silicon wafer in the regions where isolation is required. These trenches are then filled with a dielectric material, usually silicon dioxide, which acts as an insulating barrier. After filling, the wafer undergoes a Chemical Mechanical Polishing (CMP) process to remove excess dielectric and achieve a smooth, planar surface so that further fabrication steps can proceed accurately.

STI became the preferred isolation technique for advanced CMOS technologies, particularly for process nodes of 250 nanometers and below, because it allows isolation regions to be made narrower and more precise compared to earlier methods. Before STI became standard, older CMOS nodes and several non-MOS technologies typically relied on LOCOS (Local Oxidation of Silicon) for isolation.

However, LOCOS suffered from issues like the bird's beak effect, which caused lateral oxidation beyond the intended region and reduced packing density. STI overcomes these limitations by providing better control over isolation dimensions and enabling closer placement of devices. It therefore supports improved device scaling, higher packing density and reliable electrical isolation, making it an important isolation technique in modern semiconductor manufacturing.

In lithography, the STI mask pattern defines the regions where the isolation trenches are to be formed. The line-space pattern consists of alternating line and space regions within a defined boundary. Accurate dimensions and spacing of these structures are important for obtaining the desired isolation geometry and maintaining pattern fidelity during subsequent fabrication steps.

4. DESIGN PARAMETERS

Table 1: DESIGN PARAMETERS

Number	Lock	Action	Name	Material(s)	Dopant(s)	Depth	Mask	Mask Polarity	Wafer	Wafer Side	Lateral Ratio	Parameter Name	Comment
1		Wafer Setup	Wafer Setup	Silicon/Silicon...		200			WAFER_1				
2		Conformal...	pad oxide deposition	/Oxides/Silicon...		4			WAFER_1	Both	1.0		
3		Conformal...	pad nitride deposition	/Nitrides/SiN		40			WAFER_1	Top	1.0		
4		Sequence	lithography						WAFER_1				
4.1		Planarizing...	spin on resist	/Organics/photo...		120			WAFER_1	Top			
4.2		Expose resist	Expose resist	/Organics/photo...		0	RX	Light	WAFER_1	Top			
5		Sequence	etch fins						WAFER_1				
5.1		Enhanced B...	pad nitride etch	/Nitrides/Org...		40			WAFER_1	Top	0.05		
5.2		Enhanced B...	pad oxide etch	/Nitrides/Org...		4			WAFER_1	Top	0.05		
5.3		Enhanced B...	silicon etch	/Nitrides/Org...		100			WAFER_1	Top	0.05		
6		Remove M...	resist strip	/Organics/					WAFER_1				
7		Conformal...	sti fill deposition	/Oxides/SiO2_H...		200			WAFER_1	Top	0.67		
8		Polish	sti oxide cmp						WAFER_1	Top			

6. OBSERVATION

The lithography mask pattern was designed with alternating line and space regions within the defined boundary, representing the required shallow trench isolation pattern.

7. CONCLUSION

The shallow trench isolation lithography mask pattern was successfully designed and studied using Simulator3D. The line-space structures were created within the specified boundary to represent the regions required for device isolation. The experiment provided an understanding of how lithography patterns are used to define isolation regions during semiconductor fabrication.

EXPERIMENT NO. : 4DATE: 13/08/2026

1. AIM

To deposit a uniform thin film of precursor on a glass substrate using Spin Coating Technique.

2. TOOL

Spin coater

Vacuum pump

Glass substrate

Precursor/coating solution

Syringe or dropper

3. THEORY

Spin coating is a common technique used to create a thin and even layer of coating material on a flat substrate. A measured amount of precursor is placed near the centre of the substrate, which is then rotated at a controlled speed.

The process involves four key stages: deposition, spin-up, spin-off, and evaporation or film-setting. During spin-up, centrifugal force spreads the liquid outward. During spin-off, excess material is removed from the edges, reducing the film thickness.

As spinning continues, the solvent evaporates and the film becomes more stable. The final film thickness depends mainly on parameters such as spin speed, acceleration, spinning time, precursor viscosity, and solvent evaporation.

4. PROCEDURE

1. Open the lid, power on the spin coater, turn on the vacuum pump, and place the substrate on the chuck.
2. Place the substrate holder appropriately and position the properly cleaned glass substrate at the centre.
3. Switch on the spin coater and vacuum pump.
4. Check the vacuum level. It should be within approximately 250–400 mm Hg.
5. If the vacuum level is not within the required range, adjust the vacuum and check again.
6. Close the lid and run the spin-up stage.
7. Run the calibrate function.
8. Set the required parameters such as speed (rpm), acceleration (Acc.), time, and control time.
9. Dispense the required quantity of precursor/coating material onto the substrate using a syringe.
10. Run the spin-coating process.
11. After the process is completed, remove the processed sample carefully.
12. Open the lid and take out the coated substrate along with the substrate holder.
13. Close the lid, switch off the instrument, and finish the operation.

6. OBSERVATION

A thin and uniform coating of the precursor was obtained on the glass substrate after spin coating. The coating quality was influenced by the spin speed, acceleration, spinning time, vacuum condition, and quantity of precursor dispensed.

7. CONCLUSION

The spin coating process was successfully demonstrated for depositing a uniform thin film of precursor on a glass substrate. The experiment provided an understanding of how controlled rotation, vacuum holding, precursor quantity, and process parameters influence the formation and uniformity of the deposited film.

EXPERIMENT NO. : 5DATE: 13/08/2026

1. AIM

To design and study a lithography mask pattern for CMOS Inverter using the gate first approach.

2. TOOL REQUIRED

Semulator3D

3. THEORY

Lithography is a fundamental pattern-transfer process in semiconductor fabrication used to define device regions on a silicon wafer. A lithography mask contains geometrical patterns that selectively expose or protect regions of the wafer during photoresist processing. These patterns are later transferred to the underlying layers through processes such as etching and implantation. A CMOS inverter consists of complementary NMOS and PMOS transistors that provide efficient digital switching. In the gate-first approach, the gate structure is formed before the source and drain regions are defined. The mask pattern includes the active regions, well regions, gate, contacts, and metal interconnections required to form the inverter. The gate crosses the active regions to define the transistor channels, while the source, drain, and contact regions provide the required electrical connections. Using Simulator3D, the CMOS inverter mask pattern can be designed and studied layer by layer. The software helps visualize the geometrical relationship and alignment between different mask layers. Proper alignment and dimensions are essential for obtaining the intended transistor structure and ensuring correct CMOS inverter operation.

5. OBSERVATION

The lithography mask pattern for the CMOS inverter was successfully designed and studied using Simulator3D. The active, well, gate, contact, and interconnection regions required for forming the complementary NMOS and PMOS devices were observed.

6. CONCLUSION

The lithography mask pattern for the CMOS inverter was successfully designed and studied using Simulator3D. The gate-first approach helped in understanding the formation and alignment of the gate with the active regions of NMOS and PMOS devices. The experiment provided a clear understanding of how mask patterns define the structural regions required for CMOS inverter fabrication.

EXPERIMENT NO. : 6DATE: 13/08/2026

1. AIM

Demonstration of furnaces (Vacuum Oven and Muffle Furnace) for material growth

2. APPARATUS REQUIRED

Vacuum Oven

Vacuum pump

Muffle Furnace

Heating elements

Ceramic chamber or muffle

Material/sample for demonstration

Suitable sample handling tools

Personal protective equipment (PPE)

3. THEORY

A vacuum oven operates on the principle of reduced-pressure thermal desolvation. By evacuating the chamber using a vacuum pump, the pressure inside the chamber is reduced, which lowers the boiling point of moisture and trapped solvents. This allows materials to be dried at relatively low temperatures while minimizing thermal degradation.

The oxygen-depleted environment in a vacuum oven helps reduce oxidation and is useful for removing volatile solvents and moisture from heat-sensitive materials. It is commonly used for drying polymers, outgassing delicate electronic components, baking photoresists, and drying other temperature-sensitive materials.

A muffle furnace operates on the principle of isolated high-temperature indirect heating. Heating elements heat an inner ceramic chamber called the muffle, which isolates the samples from direct contact with the heating elements, flames, or combustion residues. This provides controlled and uniform high-temperature heating.

A muffle furnace is used for processes requiring high thermal energy, such as thermal decomposition, calcination, phase transformation, solid-state sintering, ashing of organic materials, metal heat treatment, and glass annealing.

Thus, the vacuum oven is mainly suitable for gentle low-temperature drying and solvent removal under reduced pressure, whereas the muffle furnace is used for high-temperature material processing and thermal treatment.

4. PROCEDURE

1. Identify the vacuum oven and muffle furnace and study their main components and controls.
2. For the vacuum oven, place the material or sample inside the chamber and ensure that the chamber is properly closed.
3. Connect the vacuum system and evacuate the chamber using the vacuum pump to obtain the required reduced pressure.
4. Set the required temperature and allow the sample to undergo controlled heating and drying under vacuum.
5. After completion, switch off the heating and allow the chamber to cool before restoring atmospheric pressure and removing the sample.
6. For the muffle furnace, place the material or sample inside the ceramic chamber using suitable handling tools.
7. Close the furnace door and set the required temperature according to the material-processing requirement.
8. Allow the furnace to heat the sample uniformly for the required duration to achieve the desired thermal treatment.
9. After completion, switch off the furnace and allow it to cool sufficiently before opening the door and removing the sample.
10. Observe and compare the operating principles and applications of the vacuum oven and muffle furnace.

6. OBSERVATION

The working and applications of the vacuum oven and muffle furnace were studied. The vacuum oven was observed to provide controlled low-temperature heating under reduced pressure, allowing efficient removal of moisture and solvents while minimizing oxidation and thermal damage.

The muffle furnace was observed to provide isolated and uniform high-temperature heating. It is suitable for thermal processes such as calcination, sintering, annealing, and thermal decomposition. The different operating principles of the two furnaces were observed based on their respective material-processing applications.

7. CONCLUSION

The demonstration of the vacuum oven and muffle furnace provided an understanding of their working principles and applications in material processing. The vacuum oven is suitable for low-temperature drying and solvent removal under reduced pressure, while the muffle furnace is suitable for high-temperature processes such as calcination, sintering, annealing, and thermal decomposition.

EXPERIMENT NO. : 7DATE: 13/08/2026

1. AIM

To design and study a lithography mask pattern for Local Oxidation of Silicon, demonstrating selective oxidation for device isolation

2. TOOL

Semulator3D

3. THEORY

Local Oxidation of Silicon (LOCOS) is a semiconductor fabrication technique used to create electrically isolated regions between adjacent devices. In this process, selected areas of a silicon wafer are oxidized to form a thick layer of silicon dioxide, while other regions are protected from oxidation. This selective oxidation is achieved using a suitable masking layer, commonly silicon nitride, which prevents oxygen from reaching the protected silicon surface. A lithography mask pattern defines the regions where oxidation is required. During the fabrication process, the mask protects the active device regions while exposing the isolation regions. Thermal oxidation is then performed, causing silicon dioxide to grow in the exposed regions. The resulting thick field oxide electrically separates neighboring active areas and reduces unwanted current flow between devices. LOCOS is an important isolation technique used in traditional integrated circuit fabrication. The geometry and dimensions of the mask pattern directly determine the location of the isolation regions and active areas. The process also demonstrates the importance of accurate lithographic pattern definition in semiconductor manufacturing. Using Simulator3D, the lithography mask pattern for LOCOS can be designed and studied to visualize the selective oxidation regions. The simulation provides an understanding of how a mask controls oxidation and how the resulting oxide regions contribute to device isolation.

4. DESIGN PARAMETERS

Table 1: DESIGN PARAMETERS

Number	Lock	Action	Name	Material(s)	Dopant(s)	Depth	Mask	Mask Polarity	Wafer	Wafer Side	Lateral Ratio	Parameter Name	Comment
1		Wafer Setup	Wafer Setup	/Silicon/Si_Xtal		400			wafer1				
2		Conformal...	pad oxide depo...	/Oxides/SiO2.T...		15			wafer1	Top	1.0		
3		Conformal...	sin hard mask d...	/Nitrides/Si3N4...		100			wafer1	Top	1.0		
4		Sequence	lithography										
4.1		Deposit	Planarizing...	/Organics/Resist		200			wafer1	Top			
4.2		Expose Mat...	UV exposure	/Organics/Resist		0	LOCOS_Trench	Dark	wafer1	Top			
5		Sequence	etch										
5.1		Enhanced B...	SiN hardmask etch	/Nitrides/, /Org...		100			wafer1	Top	0		
5.2		Enhanced B...	pad oxide etch	/Nitrides/, /Org...		15			wafer1	Top	0.3		
5.3		Remove M...	resist removal	/Organics/Resist					wafer1	Top			
6		Interface G...	thermal oxidation	/Air/, /Silicon/Si_Xtal		200			wafer1	Top			
7		Enhanced B...	SiN hardmask st...	/Nitrides/Si3N4...		105			wafer1	Top	0		
8		Multi Etch (...)	Multi Etch	/Oxides/SiO2.T...		10			wafer1	Top			

6. OBSERVATION

The lithography mask pattern for Local Oxidation of Silicon was successfully designed and studied using Simulator3D. The pattern clearly defined the regions exposed for oxidation and the protected active regions, demonstrating the principle of selective oxidation used for device isolation.

7. CONCLUSION

The lithography mask pattern for Local Oxidation of Silicon was successfully designed and studied using Simulator3D. The experiment demonstrated selective oxidation by defining protected and exposed regions through the mask pattern. It provided an understanding of how LOCOS forms field-oxide isolation between adjacent active regions during semiconductor fabrication.